# Meeting Notes

**Meeting notes moved to dedicated date tabs.**

Mar 6, 2026

Facilitator: Vince

- **Kateryna**: code for data cleanning ->, to finalize the preprocessing and feature eng notebook and setup scripts
- team review the whole current streamlit dashboard -> any improvements needed, order of pages
    - Jars - We can delete the “cookiejars page example” - i created two new pages for financial profile and previous campaign effectiveness
    - Kerensa - can delete “kerensa” page example as well - Demographic profile page has created also
- team review the README.md, update charts if needed
- kateryna please update the eda notebook, karensa and vince have added their code, suggest: use the final data: data/processed/bank-full-cleaned-processed-engineered.csv preferable, have buckets, encoded target variable, so no need to tranform varable in the final notebook, => update code in streamlit page if have time to use the final data also
- Stay active on slack for tonight to complete all tasks

Mar 5, 2026

Facilitator: Vince

Notebooks folder - Vince's suggestion

**01_eda** ->explore the raw dataset: what variables exist, distributions, missing vals, outliers, unusable vars-> **guideline** to clean, preprocess data.
**02_preprocessing** -> apply 01_eda suggestion to create data/processed/**bank-full-cleaned.csv**
**03_feature_engineering** -> add more relevant columns for analysis/models (such as add age and balance buckets, etc.) -> save to -> data/processed/**bank-full-featured.csv**

## # Business analysis

~~**04_financial_profile:** final analysis -> name your file the way you like as long as people can understand what it is about.~~
~~**05_campaign_strategy:** final analysis -> name your file the way you like as long as people can understand what it is about.~~
~~**06_previous_campaign_effectiveness:**~~
~~**07_customer_segmentation_predictions**~~
~~**08_final_report: (Optional) consolidate of best and necessary analysis -> if have time, we already have the Steamlit**~~

## # Models

**09_modeling: build model and save jotlib in to artifacts/model**
**10_eval_explainability: load model and eval.**

## # Reproducable

**Src folder:** vince will take care of this, this including scripts that can automatically download data -> clean -> add features -> train model -> save data/ models  into respective folders (replicate steps: 01 -> 03 in notebooks); so data is ready for analysis, streamlit, etc. Help stakeholders can reproduce the reports effortlessly

**Action Items**

- ☐ **Kateryna** to review 01_eda folder to ensure team's work is cohesive =  add color and formatting.
- ☐ **Vince** to complete and submitted the [README.md](README.md) file.
- ☐ **Jars** to complete slides 1 - 10, 13
- ☐ **Lanlan** to complete slides 11-12, before Saturday

**Action Items from March 4, due before March 5 meeting**

- ☐ **Kateryna** to create a 00_data_cleaning in the notebooks, and upload cleaned data into folder
- ☐ **Kateryna** will be moving all cleaned EDA to notebooks.
- ☐ **Kateryna** and Lanlan to work on model for Campaign Strategy

- [ ] **Kateryna** to create streamlit page with each of our questions: Campaign Strategy
- [x] ~~**Jarsmeka** to create streamlit page with each of our questions: Financial Profile/Previous Campaign~~
- [x] ~~**Kerensa** to create streamlit page with each of our questions: Demographics~~
- [ ] **Lanlan** and **Kerensa** to create streamlit page for Prediction Model
- [ ] **@all** to review/update DSI Showcase - March 2026
    - [ ] **Lanlan** to present on Saturday

**For visualizations**:
1) remember to make graph titles/yaxis/xaxis to be business executive/business team friendly.
2) And if there is any color needed for your visuals - lets use these:
- #1F4E52
- #2F6F73 (This one as primary)
- #5E8F95
- #8FB3C1
- #C6D9E2

Mar 4, 2026

Facilitator: Jars

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Mar 3, 2026

# MARCH 3, 2025

Facilitator: Jars

- Overall Analysis
    - Jarsmeka
        - Previous Campaign
            - Return customers are likely to resubscribe
            - Customers that said no before is less likely to say yes later
            - Recently contact customers are likely to subscribe
            - A reasonable number of follow-ups help, but too many are harmful
        - Campaigns
            - Best time to run campaigns: September to April
            - Longer phone calls (5+ minutes) increase success
            - A reasonable number (??) of follow-ups help, but too many are harmful
            - Either cell phone or telephone works
        - Demographics
            - Aim to contact individuals with jobs in:
                - blue collar
                - Management
                - Technicians
                - Admin
                - Services
                - Retirees
                - students
            - A more educated user is likely to sign up
            - Divorcees are less likely to sign up
        - Financial Profile
            - Richer clients with account balances over $1,428 are likely to subscribe - what defines a rich client
            - Clients with existing loans (personal/housing) are less likely to subscribe
            - Credit default clients are less likely to subscribe
            - Housing loans are more common than person loans
            - Aim for high-balance customers with no loans
        - **IDEAL CAMPAIGN:** The campaign should run from September to April. Focus on existing customers, especially those who previously subscribed, as prior subscription is the strongest predictor of future conversion. Target customers who were contacted recently. Use 2–3 follow-ups; more than that reduces effectiveness. Calls should last 5+ minutes, as longer duration increases success. Contact method (cell vs telephone) does not significantly impact results.
        - **IDEAL CLIENT PROFILE:** The ideal customer is financially stable, with an account balance above $1,428, no personal or housing loans, and no credit default. High-balance customers without debt have the highest conversion rates. Target individuals in blue-collar, management, technician, administrative, service roles,

retirees, and students. Higher education increases the likelihood of subscription. Divorced customers have lower conversion rates.
    - How many clients can be contacted next time based on the above profile?

**Action Items for March 3**

- [x] ~~Vince created two new buckets in the raw data~~
- [ ] Kateryna to create a 00_data_cleaning in the notebooks
- [ ] Kateryna will be moving all cleaned EDA to notebooks.
- [x] ~~Jars will be moving all cleaned EDA to notebooks.~~
- [x] ~~Kerensa will be moving all cleaned EDA to notebooks.~~
    - [ ] Format includes - Question, Code, Visualization, Analysis
    - [ ] Add blur to README to help understand organization

**Action Items for March 4, due before March 5 meeting**

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- [ ] **Kateryna** and Lanlan to work on model for Campaign Strategy
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Feb 26, 2026

# Feb 26, 2026 | DSI: Technical Learning Modules

Facilitator: Jars

Notes

- Team went over updates
- Project Plan - DSI Project Team 2
- All work should be moved to scratch folder to name: https://github.com/DzeDze/ds_team2_bank_marketing/tree/main/scratch
- Role assigned: Vince to work on Segementation EDA instead Campaign Strategy, Jarsmeka to work on showcase slides

Action items

- [x] ~~Team members will complete all analysis and visualization~~
- [ ] Team will review all pull requests/analysis before meeting again on Tuesday. The goal is to ensure we understand all findings before our next call
- [ ] Kerensa to complete changes to project proposal and submit
- [ ] Vince to create “Document” folder for README and project proposal

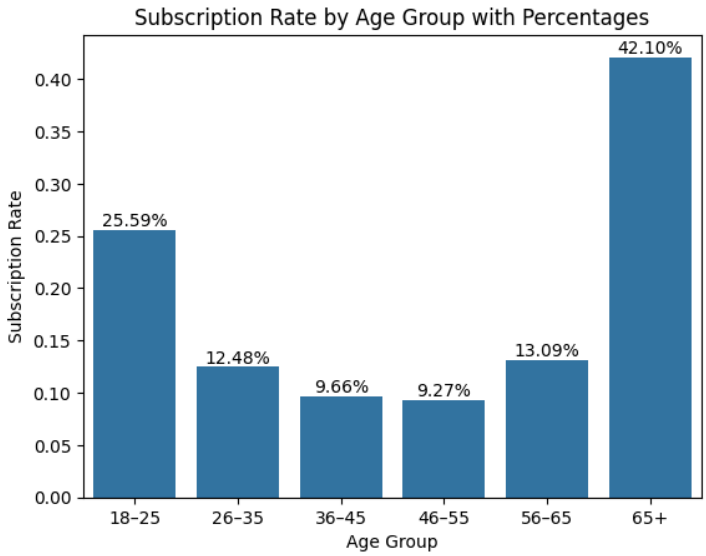
Feb 25, 2026

# Feb 25, 2026 | Team Project

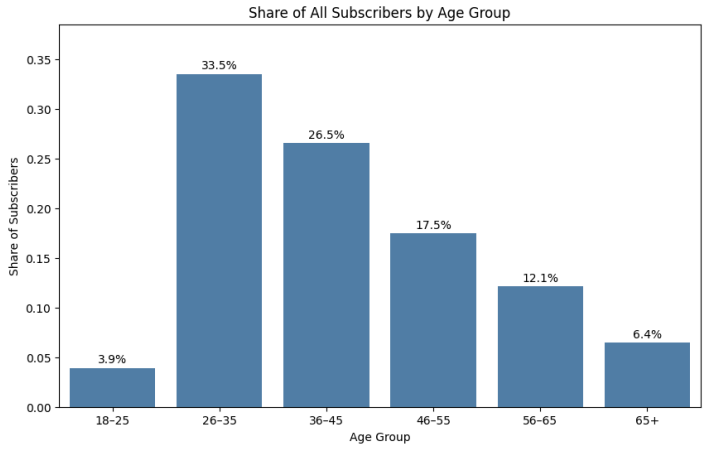
Facilitator: Jars

Notes -

- Business Problem
- Business Motivation
- Intended Audience
- Expected Impact
- <u>Compiled Questions:</u>
    a. **01) Demographics kerensa**
        i. Which age groups are most likely to subscribe to a term deposit? (Jarsmeka)




(Likelihood within each age group ← finding which groups more likely to subscribe)

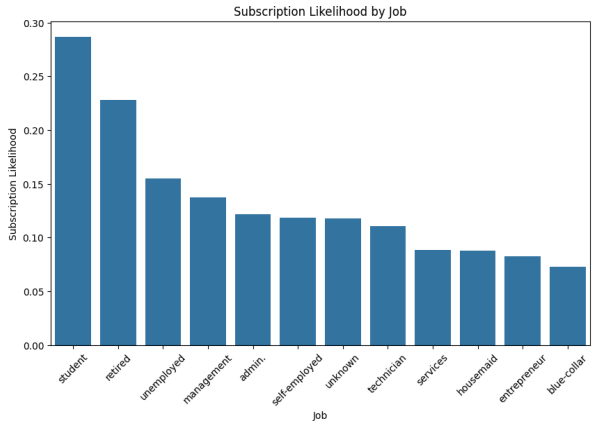



(Volume - 26-45age = most subscriber coz they are the largest group, but their subscription rate is low (not responsive) ← above graph)

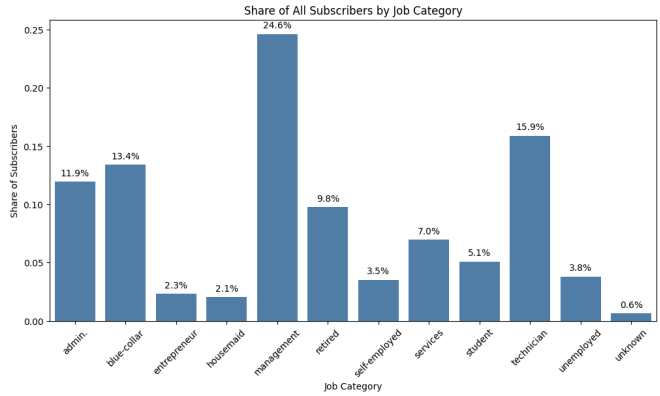
**<u>** 65+ → high conversion rate/potential; 18-25 → responsive; 26-45: large group but low responsiveness)</u>**

        ii. Are retired customers more likely to subscribe than working professionals? (Vince)

Likelihood:




Volume:

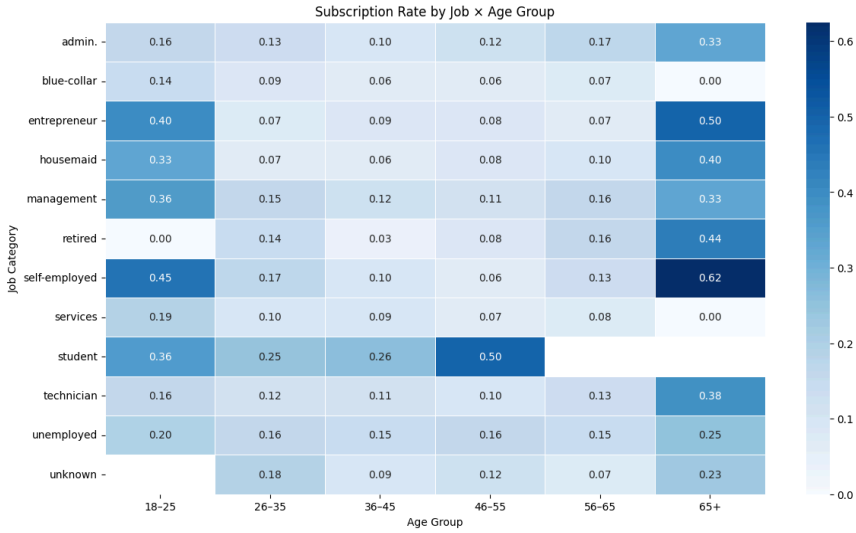



- Student and retired are easiest to convert, but they are small groups
  - Great to target, high ROI campaign
- Volume mostly come from working professionals (management, technician, blue-collar, admin)
  - Low likelihood, but large group so even small improvement in conversion can produce big gains
- High likelihood ≠ high volume (same as age group result)

iii. Does a customer's job influence subscription probability? (Vince, Jarsmeka)

Age vs job analysis

Heatmap (subscription likelihood for every job-age combination)



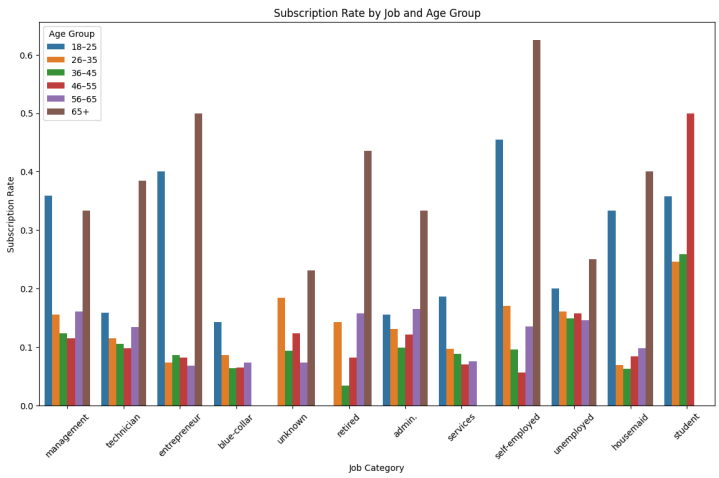

- self-employed(65+) highest subscription rate among all, retired (65+) also high
- student (18-45): high likelihood in both young and mid-age group
- blue-collar: low across all age group ← job effect dominate age effect
- management: young age and 56+ has moderate likelihood
- Entrepreneur & Housemaid: relatively high likelihood (18-25 & 65+)

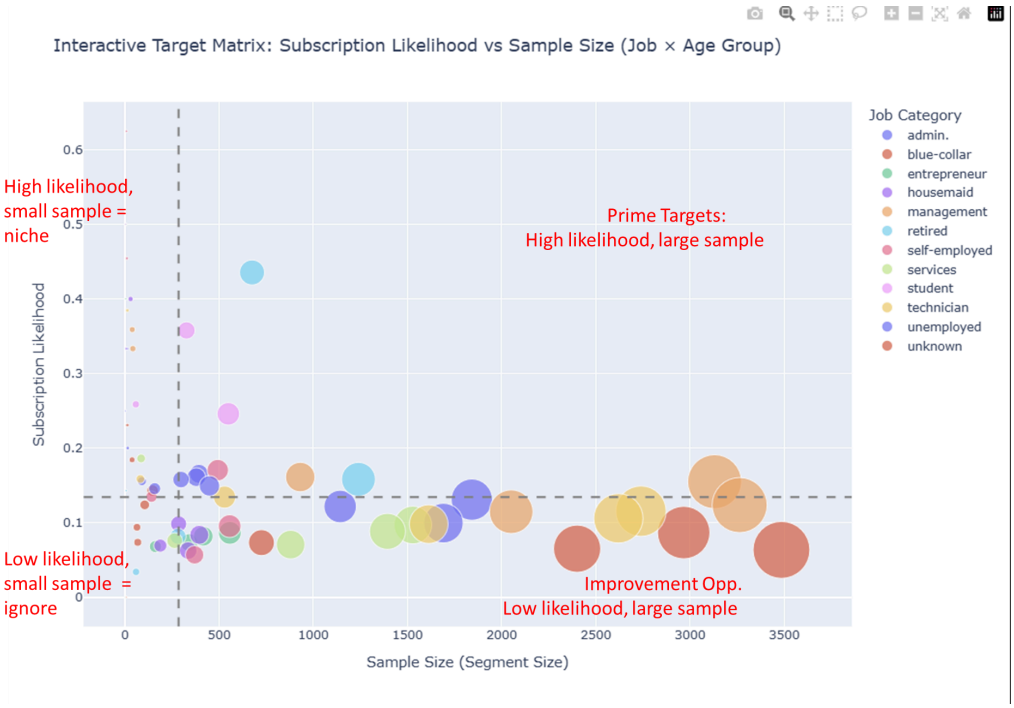
** some job (incl. Student, retired, self-employed) has high propensity

## Blue-collar (in all age) & service: low propensity **

```
  sample_size = df.pivot_table(
age_group      18–25  26–35  36–45  46–55  56–65  65+
job
admin.            90   1842   1691   1143    393   12
blue-collar      147   2967   3486   2401    725    6
entrepreneur       5    341    557    416    162    6
housemaid          6    188    336    395    285   30
management        39   3131   3264   2051    931   42
retired            1      7     59    282   1240  675
self-employed     11    493    556    370    141    8
services          86   1529   1394    880    264    1
student          327    549     58      4      0    0
technician        82   2740   2620   1613    529   13
unemployed        15    379    449    298    158    4
unknown            0     38     64    105     68   13
```

## Grouped Barplot (Volume)

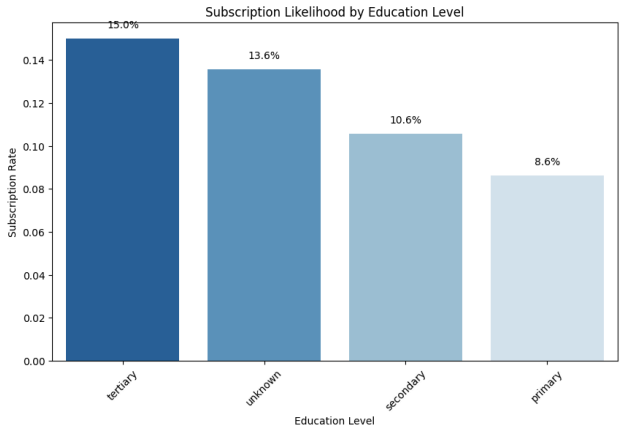




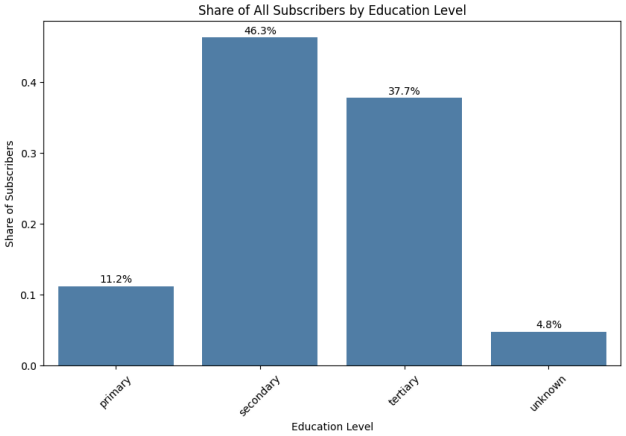

(use median to separate)

iv. Does a customer's education level impact subscription? (Jarsmeka)

Likelihood:




Volume:

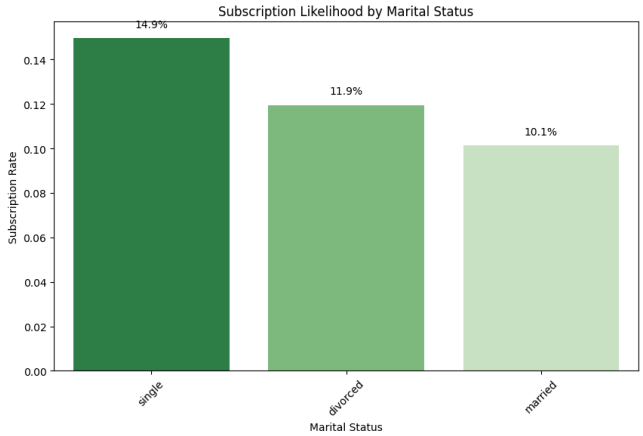


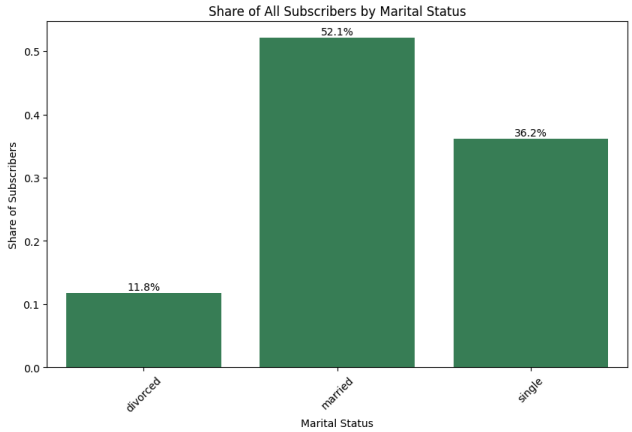

- Tertiary: highest likelihood
- Secondary: highest volume

v. How does marital status impact subscription behavior? (Jarsmeka)

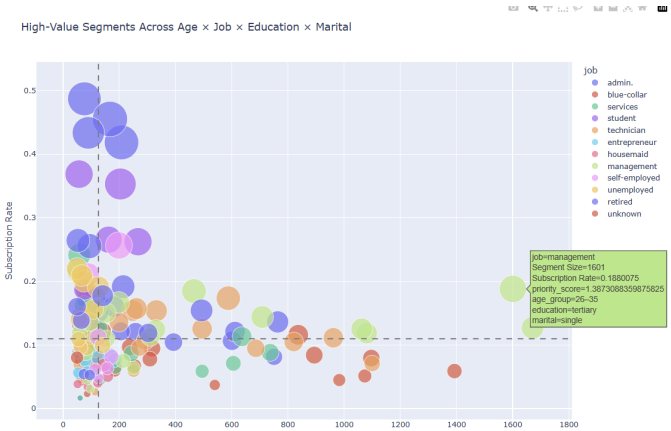
Likelihood:

Volume:




- Single: highest likelihood
- Married: highest volume




(all 4 factors)

a. **02) Financial Profile JM - COMPLETED - keep all**
   i. Does higher account balance increase likelihood of subscription? (Vince, Jarsmeka)
   ii. Does having a housing loan affect subscription probability? (Vince)
   iii. Does having a personal loan affect subscription behavior? (Vince)
   iv. Does having credit in default reduce subscription probability? (Vince)
   v. Is there an interaction between balance and loan status? (Vince)
   vi. Can subscription likelihood be linked to other products (e.g., loans)? (Kerensa)
   vii. How many loans are personal vs housing loans? (Jarsmeka)
b. **03) Campaign Strategy @Kateyna**
   i. Does contact type (cellular vs telephone) affect subscription success? (Jarsmeka)
   ii. Is there an optimal number of contacts in a campaign? (Vince)
   iii. Does subscription probability decrease after too many calls? (Vince)
   iv. What months have the highest vs lowest subscription success? (Jarsmeka, Vince)
   v. Are certain days of the week more effective? (Vince)
   vi. Is there evidence of seasonal patterns in subscription behavior? (Vince)
   vii. How does call duration relate to subscription? (Vince)
c. **04) Previous Campaign Effectiveness Jars - COMPLETED - all**
   i. How does the outcome of a previous campaign affect current success? (Vince, Jarsmeka)

ii. Are customers with previous successful campaigns more likely to subscribe again? (Vince)
iii. Are customers previously unsuccessful unlikely to convert again? (Vince)
iv. Does time since last contact (pdays) influence success probability? (Vince)
v. Are existing customers more likely to subscribe than new ones? (Kerensa)

**d. 05) Segmentation & Predictions Vince**

i. What combination of age, job, balance, and loan status yields the highest conversion rate? (Vince)
ii. Can we identify low-probability segments to avoid targeting? (Vince)
iii. Can customers be ranked by likelihood of subscription? (Vince)
iv. Does combining variables provide stronger predictive power than single variables? (Kerensa)
v. Does the combination of age + job + loan status improve prediction compared to single factors? (Kerensa)

- Risks and Unknowns

**a. Risks Kerensa**

i. The dataset may be heavily skewed toward “no” subscriptions compared to “yes” (class imbalance). (Vince, Jarsmeka)
ii. The dataset only includes campaigns from a Portuguese bank, which may limit applicability in other countries (e.g., Canada). (Kerensa, Jarsmeka)
iii. The data is older than 2013, so customer behavior and economic conditions may have changed (e.g., financial crises, COVID), reducing predictive relevance today. (Jarsmeka, Kerensa)
iv. Class imbalance may affect model accuracy and interpretation of results. (Vince)
v. Data quality concerns: “unknown” values in variables such as job, education, contact, and poutcome. (Kerensa)
vi. The “other” category in poutcome lacks explanation and may reduce interpretability. (Kerensa)
vii. Extreme outliers (e.g., very high balances or very long call durations) may skew results. (Kerensa)
viii. Predictive models may perform well on historical data but fail in real-world future campaigns. (Kerensa)

**b. Unknowns**

i. The cost of running each marketing campaign. (Jarsmeka)
ii. The profit generated per successful subscription. (Jarsmeka)
iii. Why customers choose not to subscribe (behavioral or psychological reasoning). (Jarsmeka)
How long customers maintain their term deposit after subscribing. (Jarsmeka)
iv. Whether loans are actively being used or simply held. (Jarsmeka)
v. Other non-demographic factors influencing subscription (e.g., personal motivations). (Jarsmeka)
vi. Whether the campaign strategy or call scripts changed over time. (Jarsmeka)
vii. External economic factors that may have influenced customer decisions. (Vince)

viii. Important variables not included in the dataset (e.g., income, digital engagement, geographic location). (Kerensa)
ix. Time of day of calls (morning vs evening). (Kerensa)
x. Whether customers consented to repeated contact (ethical consideration). (Kerensa)
xi. Correlation does not imply causation — observed relationships may not represent true drivers. (Kerensa)

Action items

- [ ] Analysis and Visualization to be completed by Friday
    - [ ] Initial Analysis, Demographics and Project Proposal - Kerensa
    - [ ] Financial Profile and Previous Campaign Effectiveness, Initial Showcase Slideshow - Jarsmeka
    - [ ] Campaign Effectiveness and Segmentation/Prediction - Vince